Vector Space: → collection of objects (called vectors) where we can add them together, multiply by numbers Scalars.

$\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix}$

1) $u + v \rightarrow w$

2) $u + v = v + u$

3) $u + (v + w) = (u + v) + w$

4) Zero vector doesn't change stuff. $0 + u = u + 0 = u$

5) $u + (-u) = 0$

6) ku

7) $k(u + v) = ku + kv$

8) $(k + m)u = ku + mu$

k and m → Scalar

9) $kmu = (km)u$

10) $1u = u$

Vector Space → any kind of object (no, poly, matrices, functions) as long as we can add them and multiply them by Scalars | and they follow 10 Vector Space rules.

collection of vectors where we can perform all op and all remain in the same world

1) chose set ✓ e.g 2D points, 3D points, ax^2+bx+c

2) Define + , *

$(x_1, y_1) + (x_2, y_2) =$

$(x_1 + x_2, y_1 + y_2)$

$k(x, y) = (kx, ky)$

3) check closure rule

Adding two vectors → still results in V

multiplying vector by a no → still results in V.

$(1,2), (3,4) \rightarrow (1,2) + (3,4) \rightarrow (4,6)$

Still part of V

4) check remaining 10 axioms (2-10) if all true then it is a vector space

e.g $V = \{0\}$

1) $0+0 = 0$

2) $0k \quad k = 3 \quad = 0$ hence it is a smallest possible vector space

$V = \mathbb{R}^n$ means $\rightarrow$ $(v_1, v_2, v_3, v_4 \cdots)$

$\mathbb{R}^\infty$ is a vector space of infinite list of real numbers.

Functions e.g $h(x) = 3x + 5$ $f(x) = x^2$

$F(-\infty, \infty)$ ↙

for one real no input we get real no output.

when a set is not vector space)

e.g $V = \{(x, y) \mid x > 0, y > 0\}$

$u = (3, 5)$ $ku = u$

$-1u = (-3, -5)$ x not closed

↘

we have gone outside the set

Any axiom RULE fails

then its not a vector Space

ex 8

Scalar multiplication means exponentiation

$0u = 0 \rightarrow u^0 = 1$

Subspaces → smaller set of vectors inside a vector space that still follows the same vector space

$V = \mathbb{R}^3$

e.g $W = \{(x, y, z) \in \mathbb{R}^3 \mid z = 0\}$

→ Zero vector
→ closed under add
→ closed under mult

$(1, 2, 0) + (3, 5, 0) \rightarrow$ still in W

$(2, 3, 0) \times 6 \rightarrow$ still in W

hence W subset is a subspace of V.

note: A subspace must always include a zero vector

$y = x$ ✓ $\qquad y = x + 2$ ✗

Subspaces of M_{nn} → $n \times n$ matrices

$A^T = A$ symmetric matrix

$V = M_{nn}$ → set of $n \times n$ matrices

checking if Symmetric matrices are subset

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \qquad B = \begin{bmatrix} 4 & 5 \\ 5 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} + \begin{bmatrix} 4 & 5 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 5 & 7 \\ 7 & 9 \end{bmatrix}$$

$$kA = 2\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 4 & 6 \end{bmatrix}$$ ✓

Still part of M_{nn}

Subspace

1) zero vector
2) closed under addition
3) closed under multiplication

note: Set of all continuous function forms a subspace $F(-\infty, \infty)$

$\therefore$ $a_0 + a_1 x + a_2 x^2 \cdots a_n x^n$ polynomial also closed under addition

e.g if we take exacty $n=2$

$$p(x) = 1 + 2x + 3x^2$$
$$q(x) = 5 + 7x - 3x^2$$

$p(x) + q(x) = (1+5) + 7x + 2x + 3x^2 - 3x^2$

$= (6 + 9x)$

not closed.

But if we allow all polynomials with degree $\leq n$

then it is a subspace

$P_2 \rightarrow$ Set of all poly whose degree is 2 or less.

e.g $4x + 7$, 0, $5x^2 + 3x + 1$

Buliding Subspaces:

if w_1, w_2, w_3 are subspaces of a vector space V

then their intersection is also a

subspace of V.

w_1 = all vectors on x-axis

w_2 = all vectors on y-axis

intersection.. only zero vector $(0,0)$

$V = R^2$

$w_1 = \{(x, 0)\}$ $x = 0$

$w_2 = \{(x, x)\}$ $y = x$

Insection $\rightarrow \langle 0, 0 \rangle$

$(0,0) + (0,0) \rightarrow (0,0)$

$k(0,0) \rightarrow (0,0)$

Set of all solutions to a homogeneous system $Ax=0$ is a subspace of $\mathbb{R}^n$.

↓

PROOF-

- $\langle S \rangle \rightarrow$ spanning set $\rightarrow$ (non-empty)

(Spanning sets) $w \rightarrow$ vector can be said to be a linear combination of vectors $v_1, v_2 \ldots v_r$.

$$\boxed{w = k_1 v_1 + k_2 v_2 + k_r v_2}$$

eg $v_1 = (1,0)$ $v_2 = (0,1)$ $w = (3,5)$

$3(1,0) + 5(0,1) \rightarrow (3,0) + (0,5)$

$\rightarrow (3,5)$

→ A vector space V is all possible vectors in 2D, 3D..

Set of some vectors $S = \{ w_1, w_2, w_3 \ldots \}$

$$w = \{k_1 w_1 + k_2 w_2 \cdots k_r w_r\}$$

↗

Set w of all possible combinations of the vectors in S is a subspace of V.

Spanning set → Set of all linear combinations of vectors

↳ Basically it is a set of all the linear combinations of all the vectors that are subset of vector space

notation $\langle S \rangle$

Suppose the linear combination is ↓

Linearly dependent vectors:: → $a v_1 + a_2 v_2 + a_n v_3$

$a_1 \neq 0 \quad a_2 \neq 0$.

Linearly independent vectors:: ← Otherwise they are linearly independent

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

To prove Linear Independent::

$$a v_1 + b v_2 + c v_3 + d v_4 = 0$$

$a, b, c, d = 0$

$$\left[\begin{array}{cccc|c} a & b & c & d & \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{array}\right] \rightarrow \text{using RREF}$$

then find values of a, b, c, d and if all are equal to zero then the given vectors are linearly independent

Linearly Independent and Span $\mathbb{V}$

Basis Vector → Linearly Independent.
↓
Span

let $S = \{v_1, v_2, v_3 \ldots\ldots\}$ where S is a Subspace of vector V.

Then S is said to be Basis of V

if and only Span $S = V$ and S is Linearly Independent.

then v_1, v_2, v_3 are called Basis of V.

Basis $= \mathbb{R}^3$
c_1, c_2, c_4

How to find Basis

e.g $V = \mathbb{R}^3$ $S = [v_1, v_2, v_3, v_4, v_5]$

$$a_1[1\ 0\ 1] + a_2[0\ 1\ 1] + a_3[1\ 1\ 2] + a_4[1\ 2\ 1] + a_5(-1\ 1\ -2) = [0\ 0\ 0]$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 2 & 1 & 0 \\ 1 & 1 & 2 & 1 & -2 & 0 \end{array}\right] \text{ RREF}$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{array}\right]$$

✓ ✓ ✗ ✓

now check which col has Leading 1.

Basis is c_1, c_2, c_4.

Linear combination :- let V be a vector space and $v_1, v_2 \ldots k$ be vectors in V.

A vector q can be a linear combination of $v_1, v_2 \ldots$

if

$$q = a_1 v_1 + a_2 v_2 + a_3 v_3 \ldots$$

eg $v = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}$ $v_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ $v_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$ $v_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$$a_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + a_2 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + a_3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}$$

$$\begin{aligned} 1a_1 + 1a_2 + a_3 &= 2 \\ 2a_1 + 0a_2 + a_3 &= 1 \\ a_1 + 2a_2 + 0a_3 &= 5 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 \\ 1 & 2 & 0 & 5 \end{array}\right]$$

a_1 a_2 a_3 C

NOW USE RREF to find values of a_1, a_2, a_3

eg of Span

Consider set of 2x3 matrices

$$S = \left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$$

Making a vector using these vectors.

$$q = a\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + b\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + c\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} + d\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{finalAnswer}(q) = \begin{bmatrix} a & b & 0 \\ 0 & c & d \end{bmatrix} \rightarrow \text{Result}$$

q is the linear comb of these

where a, b, c, d are real no.

So span set is

$$\text{eg} \left(\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 3 \end{bmatrix}, \begin{bmatrix} -2 & -4 & 0 \\ 0 & 6 & 7 \end{bmatrix} \ldots \ldots \right)$$

↓

these are the correct linear combinations

Q $\begin{bmatrix}1\\2\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\-1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\1\end{bmatrix}$ check whether these vectors Span R^4 or not.

$$a\begin{bmatrix}1\\2\\1\\0\end{bmatrix} + b\begin{bmatrix}1\\1\\-1\\0\end{bmatrix} + c\begin{bmatrix}0\\0\\0\\1\end{bmatrix} = \begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix}$$

converting to Augmented form

$$\left[\begin{array}{ccc|c}1 & 1 & 0 & x_1\\2 & 1 & 0 & x_2\\1 & -1 & 0 & x_3\\0 & 0 & 1 & x_4\end{array}\right]$$ use RREF

$$a = x_1 \quad b = 2x_1 - x_2 \quad c = x_4$$

Now putting these values back in a.

$$(z_2 - z_1)\begin{bmatrix}1\\2\\1\\0\end{bmatrix} + (2z_1 - z_2)\begin{bmatrix}1\\1\\-1\\0\end{bmatrix} +$$

$$z_4\begin{bmatrix}0\\0\\0\\1\end{bmatrix} = \begin{bmatrix}z_1\\z_2\\z_4\\z_4\end{bmatrix}$$

$$\begin{bmatrix}z_2 - z_1 + 2z_1 - z_2 + 0\\2z_2 - 2z_1 + 2z_1 - z_2 + 0\\z_2 - z_1 + z_1 - z_2 + 0\\0 + 0 + 0 + z_4\end{bmatrix} = \begin{bmatrix}z_1\\z_2\\z_3\\z_4\end{bmatrix}$$

$$\begin{bmatrix} z_1 \\ z_2 \\ 0 \\ z_4 \end{bmatrix} \neq \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix}$$

hence these don't Span $\mathbb{R}^4$.